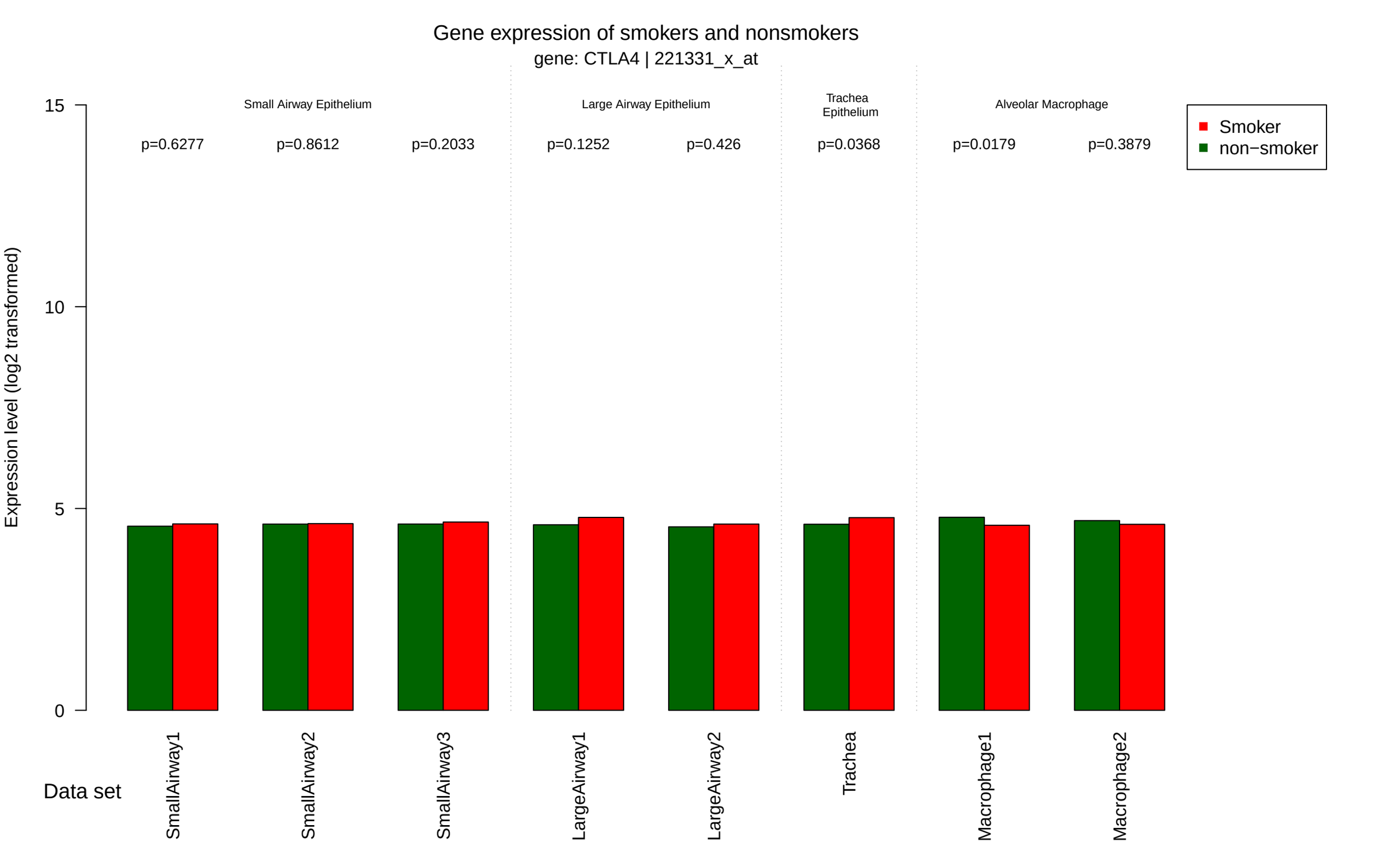
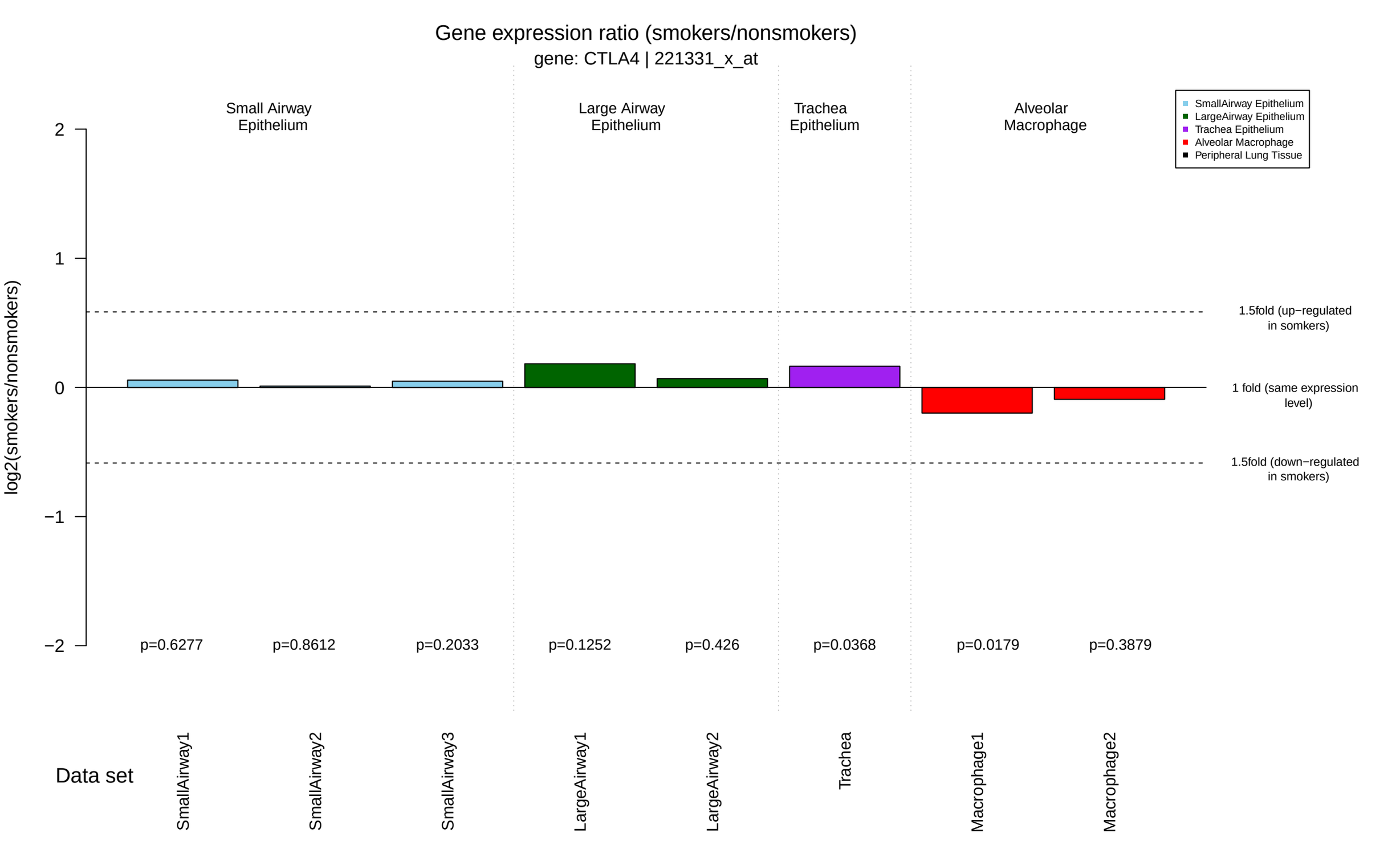
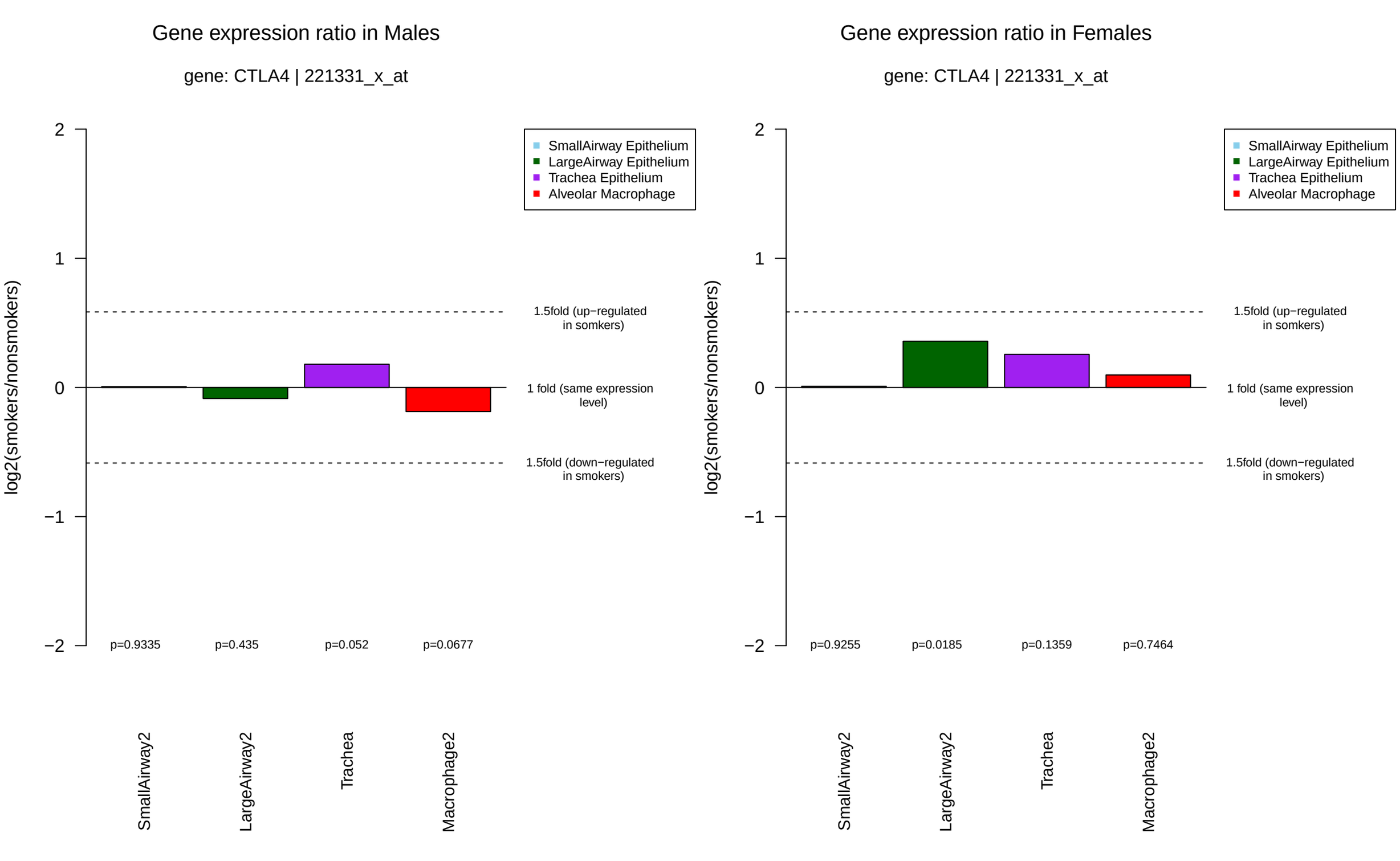
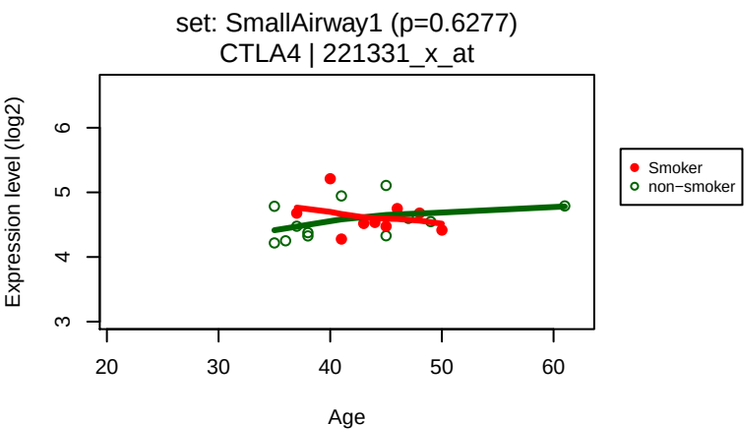
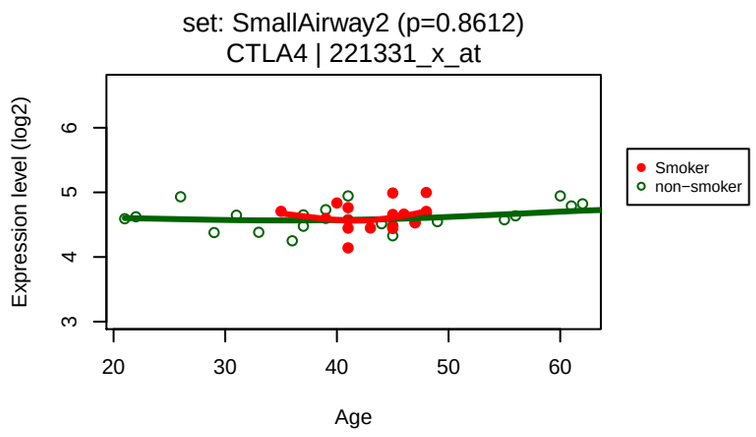
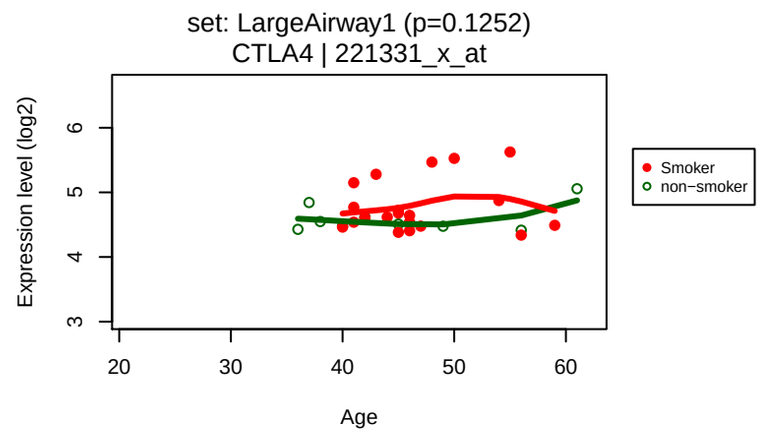

Gene expression of smokers and nonsmokers
gene: CTLA4 | 221331_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6277
p=0.8612
p=0.2033
p=0.1252
p=0.426
p=0.0368
p=0.0179
p=0.3879
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

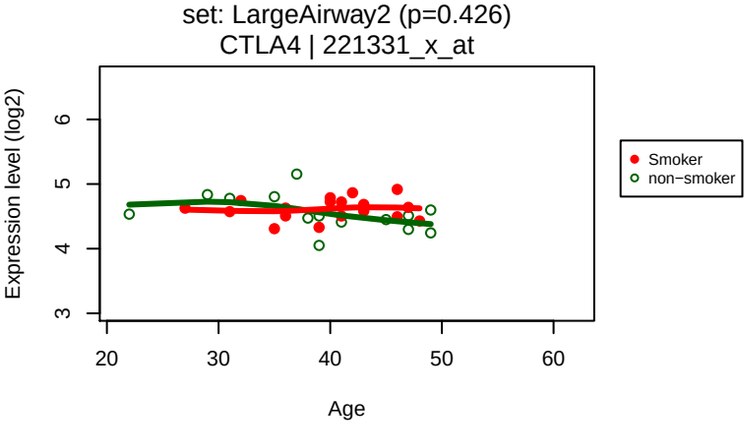

Gene expression ratio (smokers/nonsmokers)
gene: CTLA4 | 221331_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6277
p=0.8612
p=0.2033
p=0.1252
p=0.426
p=0.0368
p=0.0179
p=0.3879
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

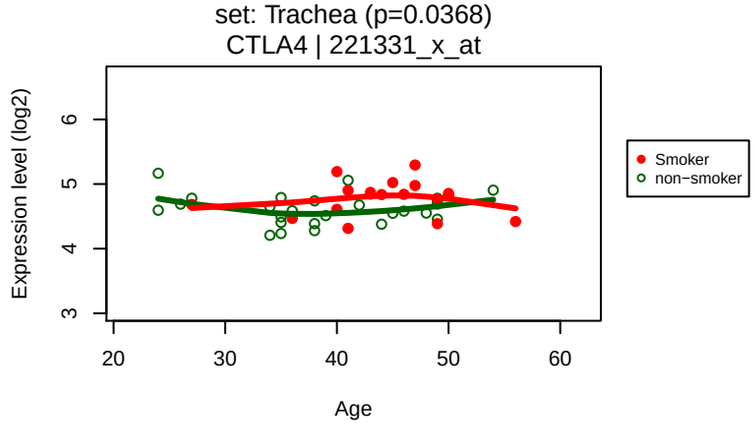

Gene expression ratio in Males
gene: CTLA4 | 221331_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9335
p=0.435
p=0.052
p=0.0677
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTLA4 | 221331_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9255
p=0.0185
p=0.1359
p=0.7464
SmallAirway2
LargeAirway2
Trachea
Macrophage2

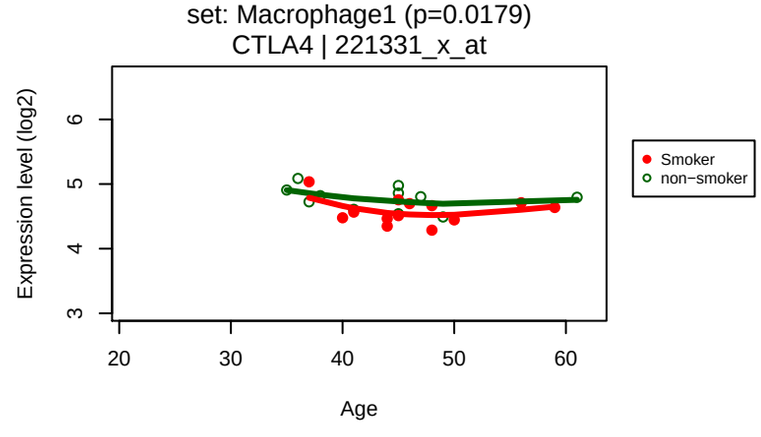

set: SmallAirway1 (p=0.6277)
CTLA4 | 221331_x_at
Expression level (log2)
Age
Smoker
non−smoker

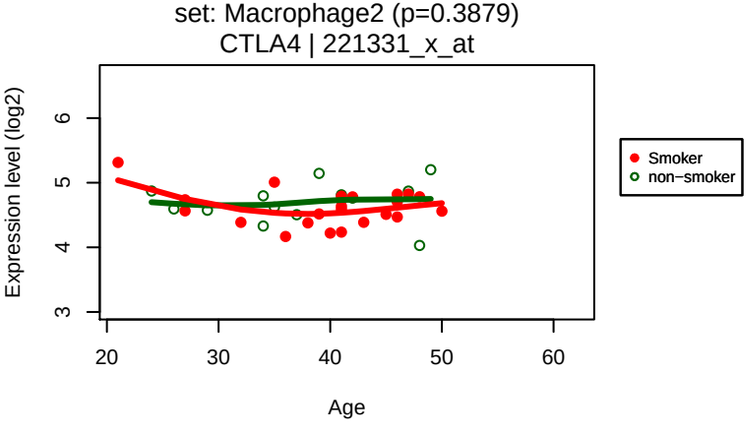

set: SmallAirway2 (p=0.8612)
CTLA4 | 221331_x_at
Expression level (log2)
Age
Smoker
non−smoker

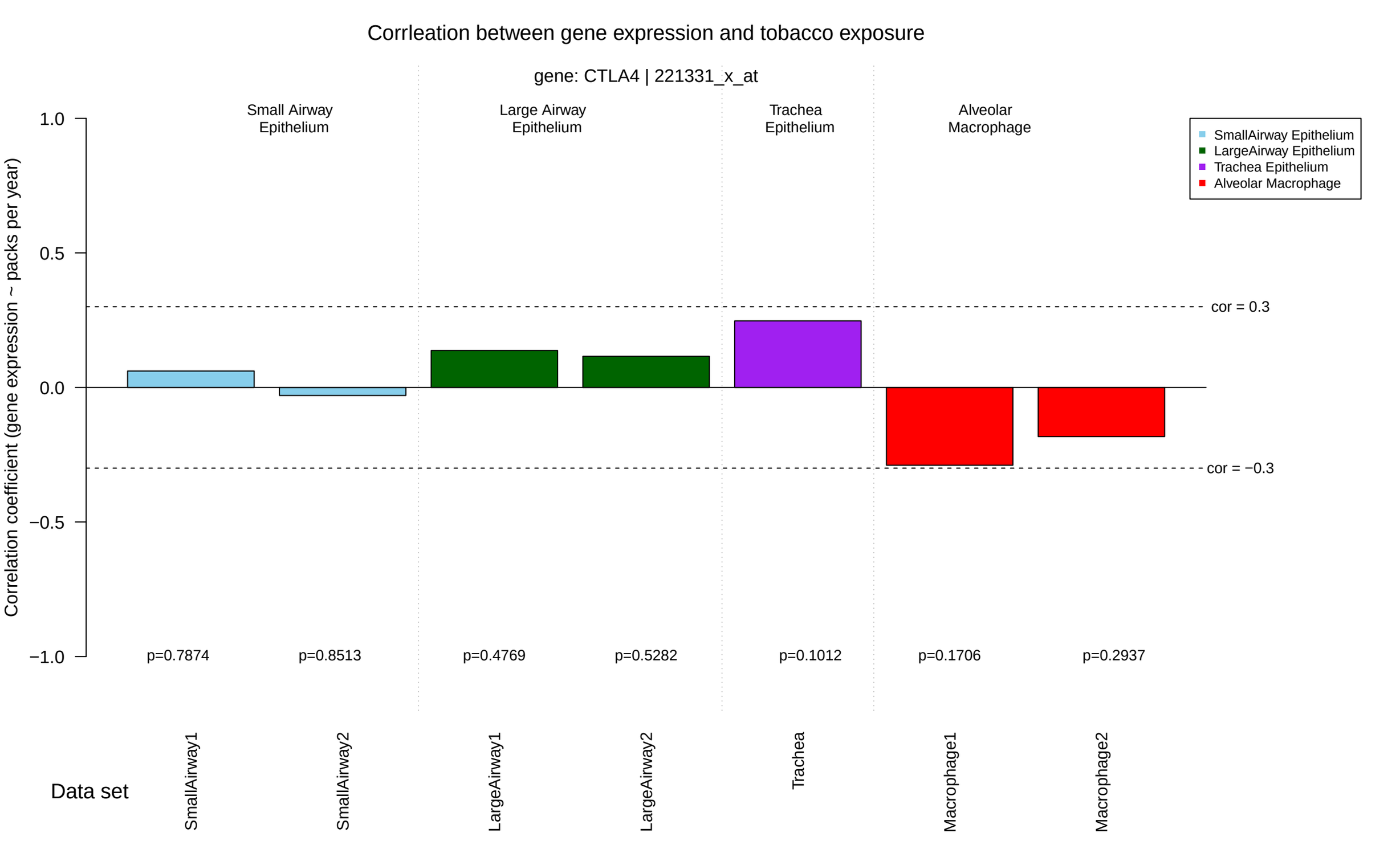

set: LargeAirway1 (p=0.1252)
CTLA4 | 221331_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.426)
CTLA4 | 221331_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0368)
CTLA4 | 221331_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0179)
CTLA4 | 221331_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3879)
CTLA4 | 221331_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTLA4 | 221331_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7874
p=0.8513
p=0.4769
p=0.5282
p=0.1012
p=0.1706
p=0.2937
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2